

Linear Algebra - Section 1.3

Vector Equations

- Def: A vector is a matrix w/ only one column. (AKA column vector).

→ We can multiply vectors by scalars.

Ex: $4 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 12 \\ -4 \end{bmatrix}$

→ We can add two vectors (if they are the same size).

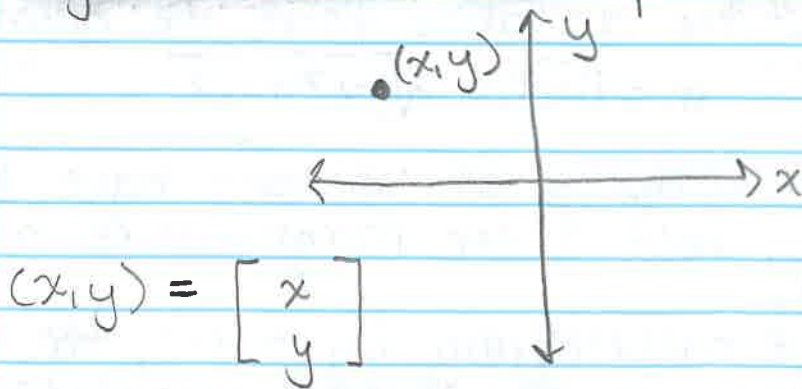
Ex: $\begin{bmatrix} 8 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 10 \\ 2 \end{bmatrix}$

→ We denote a vector by $\vec{u}$ or **u** bold face

→ Two vectors are equal if they are exactly the same.

Ex: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ vs $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ ← Not the same as the entries are not in the same places.

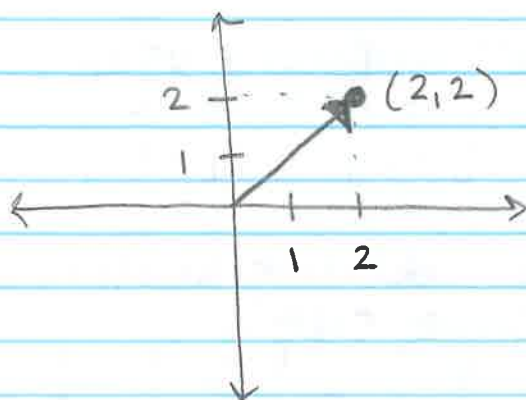
- Recall the rectangular coordinate system (Cartesian plane) ($\mathbb{R}^2$):



* we represent points in the plane as ordered pairs, but we can also view them as vectors.

- When we plot points in $\mathbb{R}^2$, we draw an arrow from $(0,0)$ to (x,y) to visually represent them as vectors.

Ex: Draw the vector $\vec{u} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$



- When viewed this way, we can see that vectors have a direction and magnitude.

→ The direction of a vector is where the vector is pointing. In 2D, direction is given by the angle it creates w/ the positive x-axis:

$$\theta = \arctan\left(\frac{y}{x}\right) \quad \text{* more complex in 3D, 4D, etc.}$$

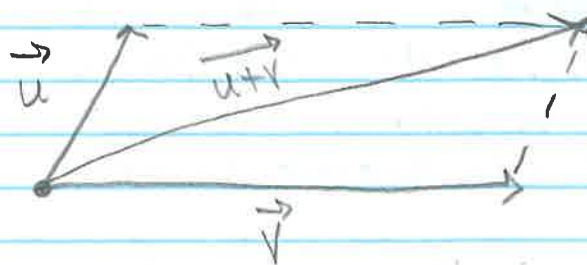
→ The magnitude of a vector is how long the vector is and is given by the distance from the origin:

$$m = \|\vec{u}\| = \sqrt{x^2 + y^2}$$

- Scalar multiples of vectors have the same direction, but their magnitudes are changed.

* If multiplying by a (-), the vector points in the exact opposite direction.

- The sum of 2 vectors has a geometric representation:



* called the parallelogram law (for obv. reasons)

- vectors in $\mathbb{R}^n$, n a natural #, are ordered n -tuples:

$$(x_1, x_2, \dots, x_n) \text{ OR } \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Algebraic Properties of $\mathbb{R}^n$:

Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be vectors in $\mathbb{R}^n$, and let c , d be scalars in $\mathbb{R}$:

- (1) $\vec{u} + \vec{v} = \vec{v} + \vec{u}$ (vector addition is commutative)
- (2) $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$ (vector addition is associative)
- (3) $\vec{u} + \vec{0} = \vec{0} + \vec{u}$ (the additive identity is the zero vector)
- (4) $\vec{u} + (-\vec{u}) = (-\vec{u}) + \vec{u} = \vec{0}$ (inverse vector)
- (5) $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$ (scalar distribution)
- (6) $(c + d)\vec{u} = c\vec{u} + d\vec{u}$ (vector distribution)
- (7) $c(d\vec{u}) = (cd)\vec{u}$
- (8) $1 \cdot \vec{u} = \vec{u}$ (scalar identity)

- Def: Given vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ and the scalars $c_1, c_2, \dots, c_n$, then the linear combination of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ w/ weights $c_1, c_2, \dots, c_n$ is defined to be:

$$c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$$

- Ex: Let $\vec{a}_1 = \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$

Is $\vec{b}$ a linear combo. of $\vec{a}_1$ and $\vec{a}_2$?

Sol: If so, then $\exists c_1, c_2 \in \mathbb{R}$ s.t.
 $c_1 \vec{a}_1 + c_2 \vec{a}_2 = \vec{b}$. That is:

$$c_1 \begin{bmatrix} 1 \\ -2 \\ -5 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} c_1 \\ -2c_1 \\ -5c_1 \end{bmatrix} + \begin{bmatrix} 2c_2 \\ 5c_2 \\ 6c_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} c_1 + 2c_2 \\ -2c_1 + 5c_2 \\ -5c_1 + 6c_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ -3 \end{bmatrix}$$

You can view it as the system:

$$\begin{cases} c_1 + 2c_2 = 7 \\ -2c_1 + 5c_2 = 4 \\ -5c_1 + 6c_2 = -3 \end{cases} \quad \leftarrow \text{we can turn this into a matrix!}$$

Yay!

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ -2 & 5 & 4 \\ -5 & 6 & -3 \end{array} \right] \leftarrow \text{Hey! wait! The columns are the same as our original vectors! } \underline{\underline{}}$$

Now do some ERO:

$$\begin{array}{l} 2R_1 + R_2 \rightarrow R_2 \\ 5R_1 + R_3 \rightarrow R_3 \end{array} \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 9 & 18 \\ 0 & 16 & 32 \end{array} \right]$$

$$\frac{1}{9}R_2 \rightarrow R_2 \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 2 \\ 0 & 16 & 32 \end{array} \right]$$

$$\begin{array}{l} -16R_2 + R_3 \rightarrow R_3 \\ -2R_2 + R_1 \rightarrow R_1 \end{array} \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right] \begin{array}{l} \leftarrow C_1 \\ \leftarrow C_2 \\ \leftarrow \text{what does} \end{array}$$

this row of 0's mean?

So, $C_1 = 3$ and $C_2 = 2$

Yes, $\vec{b}$ is a LC of $\vec{a}_1$ and $\vec{a}_2$, specifically

$$\vec{b} = 3\vec{a}_1 + 2\vec{a}_2.$$

* The row of all zeros implies that one row was redundant (we only have 2 unknowns) so, one equation was dependent on the other 2.

At most, we span a plane in 3D. "

- Lemma: $\vec{b}$ can be generated by the vectors $\vec{a}_1, \vec{a}_2, \dots, \vec{a}_n$ if and only if there is a solution to the linear system corresponding to the matrix:

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \mid \vec{b}]$$

- Def: Let $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$ be vectors in $\mathbb{R}^n$.
Then the span of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$, denoted $\text{span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r\}$, is the set of all linear combinations of $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_r$.

- Let $\vec{u}$ and $\vec{v}$ be vectors in $\mathbb{R}^n$.
 $\text{span}\{\vec{u}\}$ is a line through the origin.
 $\text{span}\{\vec{u}, \vec{v}\}$ is a plane in $\mathbb{R}^n$ that contains $\vec{u}, \vec{v}$, and $\vec{0}$.

- Ex: $\vec{a}_1 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$, $\vec{a}_2 = \begin{bmatrix} 5 \\ -13 \\ -3 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} -3 \\ 8 \\ 1 \end{bmatrix}$

Is $\vec{b}$ in $\text{span}\{\vec{a}_1, \vec{a}_2\}$?